CS 215: Task 08: Due: 30-Oct; 7:00 AM

1. In class, we derived the maximum likelihood estimator (MLE) for a normal distribution by first writing the probability density for a single observation x_i, then constructing the likelihood of the entire dataset using independence, and finally maximizing the log-likelihood.

   We now apply the same approach to a different probability model.

   Teaching-team notes. First, review the example in the book on the Bernoulli case. We are going from scalars to vectors in this question.

   In a population, a gene has two alleles, A [dominant] and a [recessive]. Every indvidual gets two copies of this gene (one from each parent) so there are three possible combinations (genotypes): AA, Aa, and aa. It is easy to see that they occur with probabilities

   $$p^2, \qquad 2p(1-p), \qquad (1-p)^2,$$

   respectively, where p is the unknown population frequency of the allele A.

   Suppose we independently sample N individuals. We observe counts of n_1 for AA, n_2 for Aa, and n_3 for aa (where $n_1 + n_2 + n_3 = N$).

   For each individual i, represent the observed genotype by a one-hot vector

   $$x_i = (x_{i1}, x_{i2}, x_{i3}),$$

   where $x_i = (1, 0, 0)$ denotes AA, $x_i = (0, 1, 0)$ denotes Aa, and $x_i = (0, 0, 1)$ denotes aa.

   (a) Write an expression for the probability $P(X_i = x_i; p)$ of a single observation, in terms of the components of x_i and the parameter p.

   Solution: For a single observation,

   $$\boxed{P(X_i = x_i; p) = (p^2)^{x_{i1}} [2p(1-p)]^{x_{i2}} [(1-p)^2]^{x_{i3}}}$$

   (b) Using the independence of the observations, construct the likelihood function $L(p)$ for the entire dataset $x_1, \ldots, x_N$.

   Solution: From independence we write

   $$L(p) = \prod_{i=1}^{N} P(X_i = x_i; p).$$

   (c) Derive the maximum likelihood estimator $\hat{p}$. Express your answer in terms of the numbers n_1, n_2, n_3 of individuals with genotypes AA, Aa, aa, respectively.

   Solution:

   1. **Define the likelihood function:**
      This is a multinomial distribution where the probabilities are constrained by a single parameter p.

      $$L(p) \propto (p^2)^{n_1} (2p(1-p))^{n_2} ((1-p)^2)^{n_3}$$

We can drop the multinomial coefficient as it does not depend on p.

$$L(p) \propto p^{2n_1} \cdot p^{n_2}(1-p)^{n_2} \cdot (1-p)^{2n_3} = p^{2n_1+n_2}(1-p)^{n_2+2n_3}$$

2. **Take the log-likelihood:**

$$\ln L(p) = (2n_1 + n_2)\ln p + (n_2 + 2n_3)\ln(1-p) + \text{constant}$$

where by constant we indicate terms not related to the minimization.

3. **Differentiate and set to zero:**

$$\frac{d \ln L(p)}{dp} = \frac{2n_1 + n_2}{p} - \frac{n_2 + 2n_3}{1-p} = 0$$

4. **Solve for p:**

$$\begin{aligned}
(1-p)(2n_1 + n_2) &= p(n_2 + 2n_3) \\
2n_1 + n_2 - p(2n_1 + n_2) &= pn_2 + 2pn_3 \\
2n_1 + n_2 &= p(2n_1 + 2n_2 + 2n_3) \\
2n_1 + n_2 &= p(2N) \\
\hat{p}_{MLE} &= \frac{2n_1 + n_2}{2N}
\end{aligned}$$

2. Let $X_1, \ldots, X_n$ be independent observations from a uniform distribution $U(0, \theta)$, where $\theta > 0$ is unknown.
   (a) Write the likelihood function $L(\theta)$, explicitly stating the values of θ for which it is nonzero.
   (b) Derive the maximum likelihood estimator of θ.
   (c) Is the estimator unbiased? Justify your answer.

   Solution:

1. **Define the likelihood function:**
   The PDF of $U(0, \theta)$ is $\frac{1}{\theta}$ for $0 \le x_i \le \theta$, and 0 otherwise.

$$L(\theta) = \prod_{i=1}^{n} \frac{1}{\theta} = \frac{1}{\theta^n}$$

Crucial step: This likelihood is only valid if *every* observed x_i is less than or equal to θ. If even a single $x_i > \theta$, the likelihood drops to 0. We express this using the maximum order statistic $X_{(n)} = \max(x_1, \ldots, x_n)$.

$$L(\theta) = \begin{cases} \frac{1}{\theta^n} & \text{if } \theta \ge X_{(n)} \\ 0 & \text{if } \theta < X_{(n)} \end{cases}$$

2. **Maximize the likelihood:**
   Calculus fails here because the function is monotonically decreasing for $\theta > 0$. If we take the derivative of θ^{-n}, we get $-n\theta^{-n-1}$, which is never zero.

   Instead, we maximize $L(\theta)$ by making the denominator θ^n as *small* as possible.

3. **Apply the constraint:**
   Since θ cannot be smaller than the largest observed serial number, the smallest possible valid value for θ is $X_{(n)}$.

$$\hat{\theta}_{MLE} = X_{(n)} = \max(X_1, X_2, \ldots, X_n)$$

3. The returns on financial assets for (volatile) cryptocurrency are sometimes modeled using a Laplace distribution, which has heavier tails (extreme jumps are more common) than a Normal distribution.

   Suppose $X_1, \ldots, X_n$ are independent observations from a Laplace distribution with probability density function

$$f(x; \mu, b) = \frac{1}{2b} \exp\left(-\frac{|x - \mu|}{b}\right),$$

   where μ is an unknown location parameter and $b > 0$ is a known scale parameter.

   (a) Write down the likelihood and log-likelihood functions for the observed data $x_1, \ldots, x_n$.

   (b) Derive the maximum likelihood estimator of μ.

   (c) Compare this estimator with the MLE of the mean of a Normal distribution. Which estimator is less sensitive to extreme observations, and why?

   Solution:

   1. **Define the likelihood function:**

$$L(\mu) = \prod_{i=1}^{n} \frac{1}{2b} \exp\left(-\frac{|x_i - \mu|}{b}\right) = \left(\frac{1}{2b}\right)^n \exp\left(-\frac{1}{b}\sum_{i=1}^{n} |x_i - \mu|\right)$$

   2. **Take the log-likelihood:**

$$\ln L(\mu) = -n \ln(2b) - \frac{1}{b}\sum_{i=1}^{n} |x_i - \mu|$$

   3. **Identify the optimization objective:**
      To maximize $\ln L(\mu)$, we must maximize the negative term. Because $b > 0$, maximizing the log-likelihood is strictly equivalent to *minimizing* the sum of absolute deviations:

$$S(\mu) = \sum_{i=1}^{n} |x_i - \mu|$$

      (Note for students: This is L_1 loss, whereas Normal distribution MLE minimizes squared error, L_2 loss).

   4. **Differentiate:**
      Standard calculus requires a slight adjustment because the absolute value function is not differentiable exactly at $x_i = \mu$. We use the signum function, denoted as $\text{sgn}(x)$, which is 1 if $x > 0$, -1 if $x < 0$, and 0 if $x = 0$.

$$\frac{d}{d\mu}|x_i - \mu| = -\text{sgn}(x_i - \mu)$$

      Taking the derivative of our sum $S(\mu)$ with respect to μ:

$$\frac{dS(\mu)}{d\mu} = -\sum_{i=1}^{n} \text{sgn}(x_i - \mu)$$

5. **Set to zero and solve:**
   To minimize the sum, we set the derivative to 0:

$$\sum_{i=1}^{n} \operatorname{sgn}(x_i - \mu) = 0$$

   Let's analyze what this equation means. The sum of the signs will be zero only if the number of positive terms equals the number of negative terms.

   In other words, the number of data points where $x_i > \mu$ must exactly equal the number of data points where $x_i < \mu$.

6. **Conclusion:**
   The value μ that splits a dataset perfectly in half (50% of the data above it, 50% below it) is the median.

$$\hat{\mu}_{MLE} = \text{Sample Median}(X_1, \ldots, X_n)$$

For a Normal distribution, the MLE for the center is the sample *mean* (minimizing squared distances). For a Laplace distribution, the MLE for the center is the sample *median* (minimizing absolute distances). Because the Laplace distribution assumes extreme outliers are more likely, its MLE naturally yields a robust statistic (the median) that ignores the magnitude of those extreme outliers.

4. A sample of 20 cigarettes is tested to determine nicotine content, and the average value observed is 1.2 mg. Suppose that the population variance is not known in advance, and the sample variance is 0.04.

   Assuming a normal population, compute a 99% two-sided confidence interval for the mean nicotine content of a cigarette.

Solution:

Since the population variance is unknown, we use the Student t-distribution.

We are given

$$\overline{x} = 1.2, \qquad n = 20, \qquad s^2 = 0.04.$$

Thus,

$$s = 0.2 \text{ and } \operatorname{se}(\overline{x}) = s/\sqrt{20}.$$

For a 99% confidence interval,

$$\alpha = 0.01, \qquad \frac{\alpha}{2} = 0.005.$$

The number of degrees of freedom is

$$n - 1 = 19.$$

From the t-table,

$$t_{0.005,19} = 2.861.$$

Therefore, the confidence interval for the population mean is

$$\overline{x} \pm t_{0.005,19} \times \operatorname{se}(\overline{x})$$

Substituting the values,

$$\overline{x} - t_{0.005,19}\frac{s}{\sqrt{n}} = 1.2 - 2.861\frac{0.2}{\sqrt{20}} \approx 1.0721,$$
$$\overline{x} + t_{0.005,19}\frac{s}{\sqrt{n}} = 1.2 + 2.861\frac{0.2}{\sqrt{20}} \approx 1.3279.$$

Hence, the 99% two-sided confidence interval is

$$\boxed{(1.0721,\ 1.3279)}.$$

5. A random sample of 1200 engineers includes 300 Hispanic American engineers, 400 African American engineers. Assume no overlap.

   Determine 90% confidence intervals

   (a) For the proportion of all engineers who are Hispanic American or African American.

   (b) The excess in the proportion of African Americans and proportion of Hispanic Americans

Solution:

For a population proportion p, an approximate $100(1-\alpha)\%$ confidence interval is

$$\widehat{p} \pm z_{\alpha/2} \times \text{se}(\widehat{p}) = \widehat{p} \pm z_{\alpha/2}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}}$$

provided the $np(1-p) > 10$ test is passed.

Here,

$$n = 1200, \qquad \alpha = 0.10, \qquad \frac{\alpha}{2} = 0.05.$$

From the standard normal table,

$$z_{0.05} = 1.645.$$

(a) The number of engineers who are Hispanic American or African American is

$$300 + 400 = 700$$

Thus,

$$\widehat{p} = \frac{700}{1200} \approx 0.58.$$

$1200\widehat{p}(1-\widehat{p}) = 292$ so the normal approximation check passes.

Therefore,

$$\widehat{p} \pm z_{0.05}\sqrt{\frac{\widehat{p}(1-\widehat{p})}{n}} = 0.58 \pm 1.645\sqrt{\frac{(0.58)(0.42)}{1200}}.$$

Hence, the 90% confidence interval is approximately

$$\boxed{(0.56,\ 0.61)}.$$

(b) For the case of excess,

$$p_H = 300/1200 = 0.25 \text{ and } p_A = 400/1200 = 0.33333$$

Again $1200p_H(1 - p_H)$ and $1200p_A * (1 - p_A)$ exceeds 10, so the normal approximation check passes.

The difference is thus $0.33333 - 0.25 = 0.08333$ and the variance for the diffence is

$$[p_A + p_H - (p_A + p_H)^2]/n = (0.58333 - 0.006944)/1200 = 0.00048032$$

so the standard error is 0.021916

The confidence interval therefore is 1.645 times the standard error

$$\boxed{(0.047,\ 0.119)}.$$

6. It is claimed that a certain type of bipolar transistor has a mean current gain of at least 210. A sample of these transistors has sample mean current gain 200 and sample standard deviation 35.

   Assuming that the population is normal, determine whether the claim should be rejected at the 5% level of significance if

   (a) the sample size is 25;
   (b) the sample size is 64.

Solution:

We test

$$H_0 : \mu \geq 210 \qquad \text{against} \qquad H_1 : \mu < 210.$$

Since the population variance is unknown, we use the statistic

$$T = \frac{\overline{X} - \mu_0}{S/\sqrt{n}}.$$

(a) For $n = 25$,

$$\overline{x} = 200, \qquad s = 35, \qquad \mu_0 = 210.$$

Thus,

$$\begin{aligned} t &= \frac{200 - 210}{35/\sqrt{25}} \\ &= \frac{-10}{7} \\ &\approx -1.429. \end{aligned}$$

The number of degrees of freedom is

$$25 - 1 = 24.$$

The p-value corresponding to the estimate is thus 0.08 which exceeds 0.05, so we fail to reject the claim at the 5% level of significance.

Equivalently, for a lower-tailed test at level $\alpha = 0.05$,

$$t_{0.05,24} \approx 1.711.$$

Hence the rejection region is

$$T < -1.711.$$

Since

$$-1.429 > -1.711,$$

we do not reject H_0.

Therefore,

$$\boxed{\text{The claim is not rejected when } n = 25.}$$

(b) For $n = 64$,

$$\begin{aligned} t &= \frac{200 - 210}{35/\sqrt{64}} \\ &= \frac{-10}{35/8} \\ &\approx -2.286. \end{aligned}$$

The number of degrees of freedom is

$$64 - 1 = 63.$$

The corresponding p-value is 0.013 which is less than 0.05. Therefore we reject the claim at the 5% signficance level.

Equivalently, for a lower-tailed test at level $\alpha = 0.05$,

$$t_{0.05,63} \approx 1.669.$$

Hence the rejection region is

$$T < -1.669.$$

Since

$$-2.286 < -1.669,$$

we reject H_0.

Therefore,

$$\boxed{\text{The claim is rejected when } n = 64.}$$

Thus, although the observed sample mean and sample standard deviation are the same in both cases, the larger sample size provides stronger evidence against the claim.